

A Low-cost Hand Recognition Based Smart Board Model (HRbSBM) for the Education System

Thiha Naing, S M Asiful Islam Saky, Md Amin Ullah Sheikh

School of Computing and Informatics,

Albukhary International University, Alor Setar, Kedah, Malaysia

thiha.naing@student.aiu.edu.my, asiful.saky@student.aiu.edu.my,

amin.sheikh@student.aiu.edu.my

Abstract: In a well growing and fast-moving organization, presenting work should be smart and emphasized looking. Proposed Model HRbSBM will provide a cozy environment to the researchers, teachers, students, and industry persons for the presentation, and it will be beneficial for school level education system where the high-cost equipment is not affordable. This proposed model (HRbSBM) will also be beneficial for those teachers who cannot stand and cannot use the white board for explanation. By using their hand gesture, they can educate the students. During the presentation, changing the presentation, drawing something on the screen, and controlling the computer is no more headache with the help of the HRbSBM. It will be able to use a hand gesture to control its computer. In the conventional method, a wireless pen or remote is provided to the presenting person and that helps to change the presentation only, while with the help of the proposed model (HRbSBM) no such gadgets are required to control it. This model not only helps during the presentation but also helps to control the computer for creating a presentation, creating graphs, and to write on screen as well. This HRbSBM will help school, college, and university students to learn in a better way and will aid teachers to present their work in a better way.

Keywords: Virtualization, Computer Vision, Smart Board, Image Processing

1. INTRODUCTION

The advent of technology has revolutionized various sectors, including education. One such promising development is the “A Low-cost Hand Recognition Based Smart Board Model (HRbSBM) for the Education System”. This research aims to explore the potential of hand recognition technology in enhancing the learning experience, making it more interactive and engaging. The HRbSBM, a cost-effective solution, leverages advanced algorithms and

techniques to recognize hand gestures, thereby enabling a more natural and intuitive way of interaction with digital educational content [18][19].

1.1. Background

Hand gesture recognition has become a significant area of research in the field of human-computer interaction [20]. Various methods have been proposed, including wearable gloves and vision-based systems. However, the former can be expensive and unstable in some environments. On the other hand, vision-based systems, like the one proposed in this research, offer a more user-friendly and cost-effective solution. Previous studies have employed pre-trained convolutional neural network models and vision transformers for hand gesture recognition [18]. This research builds upon these foundations, proposing a low-cost model specifically designed for the education system. The goal is to make digital learning more accessible and interactive, potentially benefiting students and educators alike [19].

1.2. Problem Statement:

In education system, it is mandatory for a teacher to remain stand for explanation of his/her topic, or he/she take helps the presentation. Nowadays most of the lectures are going to delivers via PowerPoint presentations by using overhead projector, which is mandatory in each school. In the light of current era technology is going to be enhanced and education system is going to switch on smart board or interactive board, but at remote or ruler areas schools cannot afford this expensive gadget, or the conventional system can be used by a normal person. But it is too difficult to an especial teacher who cannot stand and cannot write on white board easily. There is a requirement for these remote schools or especial teachers to transform knowledge with the help of the stat-of-the-art technology by using that they can be able to deliver lectures.

1.3. Research Questions:

- 1.3.1.** How a conventional white board can be transformed into a Smart Board?
- 1.3.2.** To whom this board will be beneficial?
- 1.3.3.** How will this new smart board be helpful to special teachers?

1.4. Objectives:

- 1.4.1.** A hand recognition Artificial Intelligence based model will be designed and developed.
- 1.4.2.** AI enhancement in existing conventional presentation system
- 1.4.3.** Hand gesture features will help the special teachers to demonstrate and present their ideas.

1.5. Scope & Limitations:

2. LITERATURE REVIEW

The culture of cloud computing is increasing day by day because onsite education requires a huge amount of infrastructure, and most education institutes are unable to bear the cost of such resources [1].

Due to the nature of the flexible environment of cloud computing for business, Information Technology and the availability of other required resources like automated systems and open-source software to push the users to adapt this environment. Due to its flexible infrastructure, a user can access his/her information when and where required because the working environment is not limited at home or the office [2].

In the current era the significance of cloud computing environment is habitual as well as addict the user to use the cloud services, due to its comfortable nature and user can assess his data when and where required. It is because of its fusion of intelligence and connectivity throughout the world, which enhance and change the paradigm of the network. Due to this enhancement of the nature of network education institutions which are unable to invest on infrastructure, cloud is a blessing for them. This environment leads towards the best usage of Information Technology as well as increasing the monitoring level of the quality education impart from the remote areas [3].

Due to the broader nature of cloud environment, generally it can be divided into three main categories such as: Servers, Storage capacity and Virtual environment. This environment is very useful and helpful for the all users but especially to the educational

institutions. This platform provides not only the fixability to the user but also provide the user comfort in the shape of services like google applications, moodle Edmodo or Office 365 [4].

Inculcate the education system with cloud computing environment boost up the quality of education, which provide the user maximum information and resources at minimum cost [5]. Cloud Computing cover both end the teacher and the student side, its dynamic environment facilitate the teacher to educate the student remotely which can also be economical and affordable for education sector to provide high quality education to all their students at their door step with the help of multiple tools which are available on the cloud like Office 365, Windows Google Class room etc. not only teachers and students this platform also facilitate the researchers to user the hardware as well as software which are required for their experiments and also can use the latest technology in their research. If any user required more services or extra resources, he just can bit amount and can enjoy that services [6,7,8,9,10].

According to the requirement if teacher require any specific tools which helps students to educate at on-demand bases for simulate them in his lab he can demand as per his requirement. Virtual Computers Lab is one of the good examples with pre-installed software to create quickly lab environment and students can run their code in real environment. Not only this few educational institutes are adapt the cloud services for their mail services data storage or for using Virtual Learning Environment (VLE). As education trend is going to change with the passage of time, the educational institutes are aware and also recognizing the importance the adaptability of the technology, because of it's flexible as well as efficient nature [11,12,13,14]

Other than the performance, flexible, scalable nature of the cloud computing most of the educational institutes are reluctant to use the cloud services, according to their prospective and a general survey only 4% are using cloud services, whereas among them 12% are still not aware about the cloud services, and rest of the 88% have an exploitation fear [15].

According to the literature review number of universities are going to switch on cloud computing infrastructure for educate hands-on skill to their student in real lab environment. Most of the researchers are also encourage to user remote labs or V-lab to configure their virtual machine for teach the students how to simulate and get the result from their proposed

solution. In this regard author presented a network model in R language and configure it in a virtual machine and run with mathematical and statistics to educate his students [16,17].

3. METHODOLOGY

3.1. System Requirements Specification

3.1.1. Hardware: Computer, Camera

- **Camera**

A high-resolution camera is essential for accurately detecting and tracking hand-gestures.

- Resolution: At least 1080p (1920x1080 pixels)
- Frame Rate: Minimum of 30 frames per second (fps) to ensure smooth gesture recognition
- Field of View (FOV): Wide-angle lens with a field of view between 60° to 90° to capture the full range of hand movements
- Connection Interface: USB 3.0 or higher to support high-speed data transfer

- **Computer**

The computer used for running this system must be capable of handling real-time video processing and machine-learning tasks.

- Processor (CPU): Intel Core i5 (8th generation or newer) or AMD Ryzen 5 equivalent or higher
- Graphics (GPU): Dedicated GPU with at least 4GB of VRAM, such as NVIDIA GeForce GTX 1050 or AMD Radeon RX 560. For better performance, an NVIDIA GTX 1660 or RTX 2060 is recommended.
- Memory (RAM): Minimum of 8GB RAM; 16GB RAM is recommended for smoother performance
- Storage: Solid State Drive (SSD) with at least 256GB of available space for faster data access and processing
- Operating System: Windows 10 (64-bit) or later, or Linux (Ubuntu 18.04 or later)

- Additional Peripherals: A stable internet connection for software updates and any cloud-based processing needs

3.1.2. Software:

- Importing Libraries
 - OpenCv: We used it for image processing and gesture recognition.
 - cvzone.HandTrackingModule: The HandDetector class from cvzone for hand tracking and gesture recognition.
 - numpy: The numpy library for numerical operations.
 - cv2: It is the OpenCV library for image processing.
- Initialization Video Capture

Code: `cap = cv2.VideoCapture(0)`

 - We use this to initialize the video capture from the default camera (index 0).
- Setting Video Feed Dimensions

Code: `width, height = 1500, 900`

cap.set(3, width)

cap.set(4, height)`

 - We set the desired width and height for the video feed.
- Loading Presentation Images

Code: `folderpath = "Presentation"`

pathImages = sorted(os.listdir(folderpath), key=len)`

 - We specify the folder path containing the presentation images.
 - After that, we load and sort the image filenames from the specified folder.
- Initializing Gesture Control Variables

Code: `imageNumber = 0`

buttonPressed = False

buttonCounter = 0

```

buttonDelay = 10
annotations = []
annotationsNumber = 0
annotationsStart = False`

```

- **imageNumber**: It tracks the current slide number.
- **buttonPressed**: It flag to indicate if a button has been pressed.
- **buttonCounter**: It can counter for button press delay.
- **buttonDelay**: It delays after a button press to prevent multiple detections.
- **annotations**: List of annotations for drawing on slides.
- **annotationsNumber**: Counter for the number of annotation sequences.
- **annotationsStart**: Flag to indicate the start of an annotation.

- Initializing the Hand Detector

```

Code: `detector = HandDetector(detectionCon=0.8,  

maxHands=1)`

```

- Firstly, we initialize the hand detector with a definition confidence of 0.8 and a maximum of one hand to be detected.

- Initializing Smoothing Variables

```

Code: `previousIndexFinger = None  

smoothingFactor = 0.7

```

- These variables are used to smooth the pointer movements to prevent jittering.

- Loading the Current Presentation Image

```

Code: `pathFullImage = os.path.join(folderpath,  

pathImages[imageNumber])  

imageCurrent = cv2.imread(pathFullImage)  

imageCurrent = cv2.resize(imageCurrent, (width,  

height))`

```

- We set up to load and resizes the current presentation image to fit the screen dimensions.
- Hand Detection

Code: `hands, img = detector.findHands(img)`

 - After showing the presentation slide on the screen, this code will detect our hand for further processing.
- Processing Hand Gestures

Code: `if hands and not buttonPressed:`

```

        hand = hands[0]
        fingers = detector.fingersUp(hand)
        lmList = hand['lmList']
      
```

 - After the hand detection, the system will check if hands are detected and no button is pressed.
 - Retrieves the hand landmarks and the list of fingers that are up.
- Mapping Index Finger Coordinates

Code: `xVal = int(np.interp(lmList[8][0], [width//2, w], [0, width+600]))

```

        yVal = int(np.interp(lmList[8][1], [150, height-500], [0, height]))
        indexFinger = np.array([xVal, yVal])
      
```

 - After the detection and processing, the system will map the index finger's coordinates to the presentation image dimensions.
- Button Press Handling

Code: `if buttonPressed:`

```

        buttonCounter += 1
        if buttonCounter > buttonDelay:
            buttonCounter = 0
            buttonPressed = False
      
```

 - We also implement a delay to prevent multiple detections of a single gesture.
- Drawing Annotations

Code: `cv2.line(imageCurrent, annotations[i][j] - 1, annotations[i][j], (0, 0, 200), 12)`

- After all things in above, this code will perform draws annotations on the presentation image based on the hand detections.

- **Overlaying Camera Feed**

Code: `imgSmall = cv2.resize(img, (213, 120))`

h, w, _ = imageCurrent.shape

if h > 120 and w > 213:

imageCurrent[0:120, w - 213:w] = imgSmall

- We want the image or camera of the presenter with the presentation slide on the screen. Therefore, we resize and overlay the camera feed on the presentation image.

- **Displaying Images**

Code: `cv2.imshow("Image", img)`

cv2.imshow("Slides", imageCurrent)`

- This will display the camera feed and the current slide with annotations.

- **Exit Condition**

Code: `key = cv2.waitKey(1)`

if key == ord('q'):

break

cap.release()

cv2.destroyAllWindows()`

- If the user wants to exit from the system, he or she needs to press 'q' key on the keyboard to exit from the system.
- Releases the video capture and closes all OpenCV windows.

3.2. Model Selection

For our hand gesture presentation system, we picked a model provided by the `cvzone` library, which is built upon Python version 3.12. Our development environment also utilized a laptop Windows 11, equipped with a built-in webcam. We decided to use this model based on several factors, including its compatibility with our Python environment, ease of integration with our existing codebase, and ability to effectively detect and track hand gestures in real time.

The model utilized for hand gesture recognition within our system is part of the `cvzone` library, specifically the HandTrackingModule. This module incorporates machine learning algorithms that are trained on hand gesture data to accurately detect and track hand movements in real-time.

- 3.2.1. **Compatibility:** The chosen model is compatible with Python 3.12, our preferred programming language version, and ensures seamless integration into our development environment.
- 3.2.2. **Operating System Compatibility:** Given that our development environment runs on Windows 11 compatibility with this operating system was a crucial consideration.
- 3.2.3. **Hardware Compatibility:** The model's compatibility with the built-in webcam of our laptop ensures optimal performance without the need for additional hardware.
- 3.2.4. **Real-Time Performance:** While the model shows a slight latency, with an average response time of 5 to 6 seconds, it still meets our requirements for real-time hand gesture recognition in the context of controlling presentations.
- 3.2.5. **Accuracy:** Regardless of the latency, the model demonstrates satisfactory accuracy in detecting and tracking hand gestures, ensuring reliable interaction with presentation slides.

3.3. System Integration

In this section, we hunt through the seamless integration of real-time hand gesture recognition into our presentation control system and cover the intricacies of gesture detection, interpretation, and control mechanisms.

3.3.1. Real-time Gesture Recognition

Our system harnesses the power of the `cvzone` library's HandTrackingModule to achieve real-time detection and tracking of hand gestures. This module employs machine learning algorithms that is trained on extensive hand gestures to accurately identify and monitor hand movements within the webcam feed. By analyzing key landmarks such as finger orientations, the system effectively interprets various hand gestures in real time.

**Code: `detector = HandDetector(detectionCon=0.8, maxHands=1)
hands, img = detector.findHands(img)`**

The HandTrackingModule operates by first detecting the presence of hands within the webcam frame. Once detected, the module proceeds to track the movement of each hand, continuously updating the position of key landmarks. This dynamic tracking enables the system to capture precise hand gestures and translate them into meaningful actions within the presentation environment.

3.3.2. Control Mechanism

Our system offers an adaptable control mechanism that seamlessly integrates with the detected hand movements or gestures, providing users with instinctive control over presentation navigation and annotation. Key gestures such as extending the index finger to point or making swiping motions for slide transitions, are mapped to specific functions within the presentation software.

**Code: `if fingers == [1, 0, 0, 0, 0]:
print("Previous slide (left)")
imageNumber -= 1`**



Figure 1: Thumbs up

```
if fingers == [0, 0, 0, 0, 1]:  
    print("Next slide (right)")  
    imageNumber += 1
```



Figure 2: Pinky Finger up

```
if fingers == [0, 1, 1, 0, 0]:  
    cv2.circle(imageCurrent, indexFingerTuple, 10, (0, 0,  
255), cv2.FILLED)
```



Figure 3: The Index Finger and Middle Finger up

```
if fingers == [0, 1, 0, 0, 0]:  
    cv2.circle(imageCurrent, indexFingerTuple, 10, (0, 0,  
255), cv2.FILLED)
```




Figure 4: The Index Finger up

```
if fingers == [0, 1, 1, 1, 0]:
    annotationsNumber -= 1`
```



Figure 5: The Index Finger, Middle Finger and Ring Finger up

For instance, extending the index finger (Figure 4) allows users to point to specific areas of the presentation slide, providing a natural and intuitive means of highlighting key points or directing the audience's attention. Similarly, performing a swiping gesture in a specified direction triggers the navigation to the next or previous slide, facilitating smooth transitions between presentation content. Extending both the Index finger and Middle finger (Figure 3) will show a pointer on the slide and the presenter can play around with the pointer to point out the information on the slide that he or she wants to show. Extending the Pinky finger (Figure 2) will perform a moving slide forward and the Thumb up (Figure 1) will perform a moving slide backward. The last one is extending the Index finger, Middle finger and Ring finger together (Figure 5) will erase the annotations that we draw on the slide.

The integration process involves establishing robust communication between the HandTrackingModule and our presentation control software. This involved developing an interface to

capture and interpret detected hand gestures, mapping them to corresponding actions within the presentation environment. Despite the slight latency observed, with an average response time of 5 to 6 seconds, the system delivers satisfactory performance, ensuring smooth and responsive interaction with presentation content.

3.4. Testing and Evaluation

In this section, we will deliver the comprehensive testing and evaluation process conducted to assess the performance, reliability, and usability of our real-time hand gesture recognition system for presentation control.

3.4.1. Test Scenarios

Our testing strategy encompassed a wide range of scenarios to ensure through evaluation of the system capabilities:

- **Lighting Conditions:** We tested the system under various lighting conditions, including bright, dim, and artificial lighting setups, to assess its robustness and adaptability to different environments. The system can detect and track the hand well and the performance of our system is going well with varying degrees of lighting.
- **Hand Orientations:** The system was evaluated for its ability to detect and track hand gestures across different orientations, such as frontal, side, and overhead views, to ensure consistent performance regardless of hand position. But the hand needs to be in the range of the webcam frame and the system can not detect it when the hand is out of frame or boundary.

Code: `width, height = 1500, 900

cap.set(3, width)

cap.set(4, height)`

- **Distance from Camera:** We tested the system's performance at different distances from the webcam, ranging from close proximity to several meters away, to determine its optimal operating range and accuracy. While we are testing our system, our system is working well with a distance between 30cm to 70 cm. Lower than 30cm, the hand is really close to the camera and the system can not detect it well and also, when we put the hand more than 70cm apart from the camera, the hand is really far away from the camera and the system can not track the hand.
- **Gesture Complexity:** We used various gestures of differing complexity were evaluate, ranging from simple pointing gestures to more intricate drawing and multi-finger gestures, to measure the system's versatility and precision. Also, our system allows only one hand while using the system in order to terminate the gesture complexity.

Code: `detector    =    HandDetector(detectionCon=0.8, maxHands=1)`

- **Simple Gestures:** Simple gestures typically involve straightforward hand movements that can be easily recognized and interpreted by the system. Examples include:
 - **Annotation (Drawing):** Draw a motion with the index finger `[0,1,0,0,0]` to highlight or annotate a specific area of the presentation slide.
 - **Moving Forward:** Make a swipe with the little finger `[0,0,0,0,1]` to move the slide to right or forward.
 - **Moving Backward:** Make a swipe with the thumb `[1,0,0,0,0]` to move the slide to the left or backward.

- **Multi-finger Gestures:** Multi-finger gestures involve coordinated movements of multiple fingers and are often more intricate than simple gestures. Examples include:
 - **Pointing:** Extending the index finger and middle finger `[0,1,1,0,0]` to indicate a specific location or direction on the slide we want to point out.
 - **Erasing:** Extending the index finger, middle finger, and ring finger `[0,1,1,1,0]` to erase everything we draw on the slide in a sequence.

3.4.2. Performance Metrics

To quantitatively assess the performance of our gesture recognition system, we employed the following metrics:

- **Accuracy:** The percentage of correctly detected hand gestures relative to the total number of gestures performed, providing insight into the system's reliability in accurately interpreting user input. Our system's accuracy for finger detection or hand gesture detection is working well and it has an accuracy of 100%.
- **Latency:** The average time taken for our system to detect and respond to a hand gesture, measured in milliseconds or seconds, to evaluate its responsiveness and real-time performance.

3.4.3. User Testing

In addition to quantitative metrics, we conducted extensive user testing sessions to gather qualitative feedback on the usability and user experience of the gesture recognition system.

- **Task Completion:** Users were tasked with performing common presentation control actions, such as navigating slides, annotating content, and activating features using gesture-based controls. Their ability to complete these tasks efficiently and effectively was assessed. In our system, we already tested all

these things that deal with hand gestures, and these all abilities are working well in the user testing stages.

- **Usability Issues:** Any usability issues, challenges, or limitations encountered by users during interaction with the system were documented and analyzed. These insights were used to inform iterative development and optimization efforts. While we were testing the system, we faced some issues such as the pointer not working properly and when drawing with the index finger on the slide, the line we drew was not coming on the slide also moving the slide was not working properly sometimes it is too fast and sometimes it is really slow and it takes a long time to move to the next slide.

By testing the system across diverse scenarios, measuring key performance metrics, and soliciting user feedback, we were able to comprehensively evaluate the effectiveness and usability of our real-time gesture recognition system for presentation control.

3.5. Flow of the System

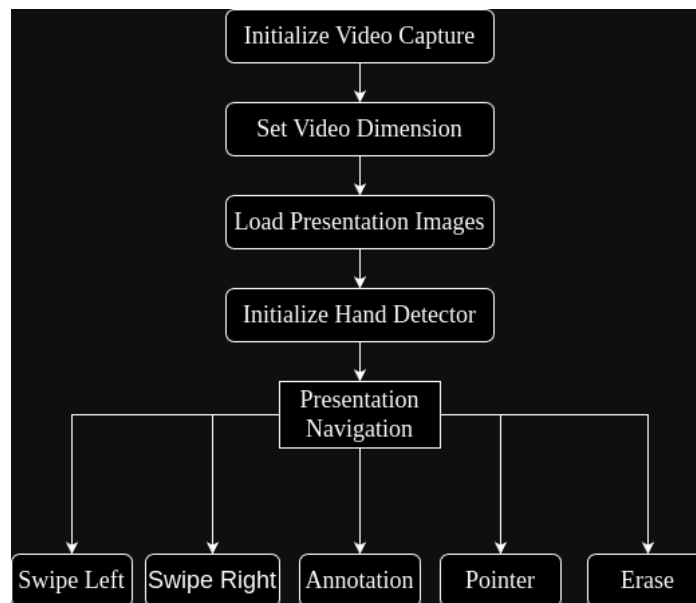


Figure 6: Flow of our system (A Low-cost Hand Recognition Based Smart Board Model (HRbSBM) for the Education System)

- 3.5.1. Initialize Video Capture:** The first step is initialization video capture and this step involves the video capture process using a webcam. The webcam will be used to detect and track hand movements. This enables real-time hand gesture recognition by capturing live video feeds.
- 3.5.2. Set Video Dimension:** The second step is setting the video dimensions and the dimensions of the video feed are set to ensure that the captured video is of appropriate size and resolution for accurate hand detection. This standardizes the video input for consistent and reliable gesture recognition.
- 3.5.3. Load Presentation Images:** This step is the third one and the presentation images or slides are loaded into the system. These images will be displayed during the presentation.
- 3.5.4. Initialize Hand Detector:** This step involves initializing the hand detection algorithm. The system sets up the necessary parameters and models to recognize hand gestures from the video feed.
- 3.5.5. Presentation Navigation:** This step involves the core functionality of our system where the recognized hand gestures are mapped to specific actions for navigating and interacting with the presentation. The purpose of this step is to control the presentation and perform various functions based on the hand gestures.
- **Swipe Left:** This gesture is used to navigate to the previous slide in the presentation.
 - **Swipe Right:** This gesture is used to navigate to the next slide in the presentation.
 - **Annotation:** This gesture allows the parameter to draw or write annotations on the current slide.
 - **Pointer:** This gesture activates a pointer on the slide and allowing the presenter to highlight specific areas of the slide that he or she want to point out.
 - **Erase:** This gesture is used to erase any annotations or markings that the presenter makes on the slide.
- 3.5.6. A**
- 3.5.7.**

4. RESULTS & ANALYSIS

4.1 Presentation Navigation and Annotation

4.1.1 Swipe left : For moving to the previous slide, the hand gesture should be made by first finger. The thumbs up hand gesture(Figure 7: Swipe Left with Thumb) will move the slides backward.



Figure 7: Swipe Left with Thumb

4.1.2 Swipe right: Moving the slides forward, executed with the last finger or 5th finger(pinky finger). The pinky finger gesture(Figure 8: Swipe Right with the Pinky finger) will move the slides to the next one.

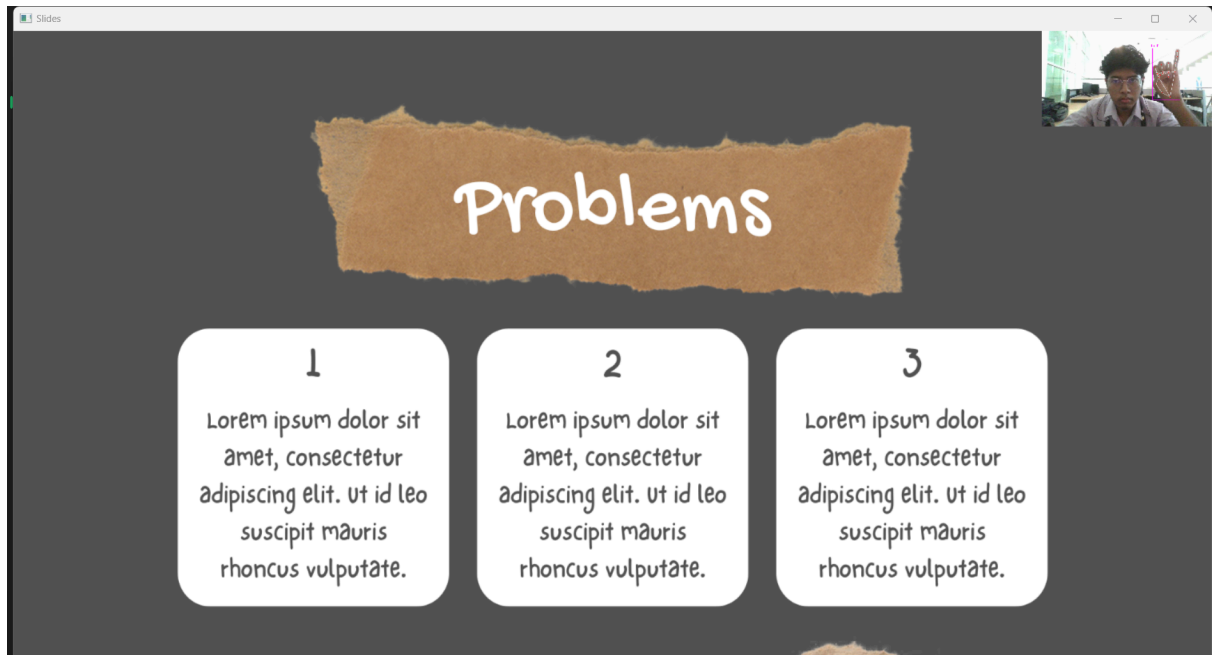


Figure 8: Swipe Right with the Pinky finger

4.1.3 Annotation: Annotation is also possible with this system by doing index finger gestures(Figure 9: Annotation with the Index finger). Moving the index finger accordingly with respect to the position of expected annotation areas will draw annotations in the slide.



Figure 9: Annotation with the Index finger

4.1.4 Pointer: A red dot pointer will appear on the screen with two fingers gesture. With the index finger and middle finger gesture, the red dot will appear on the screen and moving these two fingers together will change the pointer position to navigate the explanation points accurately.



Figure 10: Pointing with both Index finger and Middle finger

4.1.5 Erasing annotations: Once annotations are made, can be erased by three fingers gesture(Figure 11: Erasing Annotations with Three Fingers(Index finger, Middle finger and Ring finger together)). The index finger, middle finger and ring finger, together with these three fingers gesture and pointing to the exact annotated areas, annotations can be erased by moving these three fingers.



Figure 11: Erasing Annotations with Three Fingers(Index finger, Middle finger and Ring finger together)

Presentation of ideas is one of the most important things in education and business. While presenting, the presenter may find it interrupting to change the slides and pointing manually. Here, the system proposes the solution by minimizing the interruptions and allowing presenters to focus more on their content and audience engagement. They just simply can make the slide transitions and annotations using hand gestures. The different unique hand gestures will allow them to move back to the previous slide, to move forward and also the annotation feature allows users to switch easily into annotation mode to draw or write on slides effortlessly. The most important thing is pointing the exact line or content by hand gestures rather than using a laser pointer, that will keep the flow of the presentation. However, some challenges were found in maintaining continuous drawing performance under different lighting conditions. But our system has a higher accuracy rate with well responsiveness to enhance the user experience in presenting by having their audience's attention directly and effectively, making the presentations more engaging and informative.

5. CONCLUSION

The research on “A Low-cost Hand Recognition Based Smart Board Model (HRbSBM) for the Education System” has shed light on the transformative potential of hand recognition technology in the realm of education. The HRbSBM, with its cost-effective and user-friendly

approach, has the potential to revolutionize the way educational content is interacted with, making learning more engaging and interactive. This research has also highlighted the importance of making such advanced technology accessible to remote schools and special teachers, thereby democratizing the benefits of digital learning. The proposed AI-based model, designed to enhance the existing conventional presentation system, is expected to be a significant step towards this goal. By enabling special teachers to demonstrate and present their ideas through hand gestures, this research contributes to the ongoing efforts to make education more inclusive and effective. As we continue to explore and refine this technology, we look forward to witnessing its positive impact on the education system.

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3. Methodology

3.1 System Requirements Specification

Hardware: Computer, Camera

Software Part: IDE, Libraries: OpenCV, MediaPipe, Pandas, Numpy

3.2 Model Selection

For real-time gesture recognition in your system, you've chosen a model provided by the cvzone library. This model is likely based on machine learning algorithms trained on hand gesture data to accurately detect and track hand movements in real time. You can mention the specific model used for hand tracking and the rationale behind selecting it for your system. Discuss factors such as accuracy, speed, and ease of integration with your existing codebase.

For our hand gesture presentation system, we have chosen a model provided by the `cvzone` library. This model is likely based on the machine learning algorithms that is trained on hand gesture data to accurately predict, detect, and track hand movements in real time.

3.3 System Integration

Real-time Gesture Recognition

Control Mechanism

3.4 Testing and Evaluation

Test Scenarios

Performance Metrics

User Testing

(Flowchart)

4. Results and Discussion

4.1 Navigations

Right:

Left

Drawing/Annotation

Pointer

Erasing

4.2 Performance metrics and Accuracy of Gesture Recognition

Response Time

Gesture Recognition Rate

Annotation Accuracy

Discussion

4.3 Interpretation of Results

4.4 Challenges and Limitations

Draft

Future Work

Improvements and Enhancements

Several improvements and enhancements can be made to the current system:

Gesture Set Expansion: Expanding the gesture set to include more commands and functionalities.

Adaptive Algorithms: Developing adaptive algorithms that can adjust to different lighting conditions and background complexities.

User Customization: Allowing users to customize gestures according to their preferences.

Further Research

Further research can explore the integration of this gesture-controlled system with other interactive technologies:

Virtual and Augmented Reality: Combining gesture control with VR and AR environments to create immersive and interactive experiences.

Collaborative Tools: Enhancing remote collaboration tools by integrating gesture controls for better interaction and engagement.